FocusLens

A simple project explanation

What is FocusLens?

FocusLens is a simple desktop application that helps a person understand how they use their computer. It watches the application currently open on the screen and saves the time spent on it. The app then shows whether that time was productive, neutral, or distracting. The main purpose is to help students, professionals, and anyone who wants to build better focus habits.

How does it work?

When the user starts tracking, FocusLens checks the active Windows application after a short interval. For example, coding tools, Word, Excel, GitHub, and learning sites can be treated as productive. YouTube, Instagram, Netflix, and similar apps or sites can be treated as distracting. The program stores only local usage summaries in an SQLite database on the computer. It does not need a website, cloud account, or internet connection.

Main dashboard

The dashboard gives a quick view of the day. It shows a focus score, productive time, distracting time, app switches, and focus efficiency. Pie and bar charts make the information easy to understand. The daily goal bar compares productive minutes with the target selected by the user. This gives a clear answer to a simple question: am I using my time in a useful way today?

Useful new features

Users can add their own app categories in Settings. For example, a user may mark Slack as productive for work, while another user may mark it as distracting. Users can also choose a daily productive-time goal. The Deep Focus page has a Pomodoro timer where work and break lengths can be changed. Completed focus sessions are saved as a small history. A gentle desktop reminder appears when a distracting app is used during Focus Sentinel mode.

Reports and privacy

The Weekly Report page gives a simple summary of the last seven days. It shows the focus score, productive time, the biggest distraction, the best focus hour, and one useful improvement tip. Data can also be exported as a CSV file. The user can clear their saved history whenever they want. Because the database stays on the same computer, the project keeps the user in control of their information.

Technology used

FocusLens is made only with Python. CustomTkinter and Tkinter are used for the desktop interface. SQLite is used to save data locally. Psutil and Windows system functions help identify the active application. Pandas and Matplotlib are used to calculate results and draw charts. The startup option uses the Windows user startup setting, so the app can open automatically after sign-in if the user chooses it.

In simple words

FocusLens is like a personal study and work companion. It does not block the user or judge them. It simply shows where time goes, reminds them when attention slips, and helps them improve one day at a time. It is useful for people who want to study with fewer distractions, finish tasks on time, understand their digital habits, and make small improvements every day.